PHASE 7 DOCUMENT

ComicCraft: AI Comic Story Creator using Gemini Models

Phase Name: System Testing, Quality Assurance and Security Validation

Project Name	ComicCraft: AI Comic Story Creator using Gemini Models
Phase	Phase 7 – Testing, QA and Security Validation
Project Type	Generative AI Comic Creation Application
Main Technology	Google Gemini Models
Prepared By	_____
Guide / Faculty	_____
Date	_____

Academic Project Phase Documentation

1. Phase Overview

This is Phase 7 of **ComicCraft: AI Comic Story Creator using Gemini Models**. This phase evaluates the complete system after the core generation, editing, preview, export, privacy, safety, and collaboration features have been developed. It covers functional testing, integration testing, usability review, content-safety validation, performance checks, security testing, defect management, and user acceptance preparation.

2. Phase Objective

The objective of this phase is to verify that ComicCraft meets its requirements and provides a dependable, safe, and understandable AI-assisted comic-creation experience. Testing confirms that users can create, edit, save, preview, export, share, and manage comic projects correctly, while Gemini integration, account controls, privacy settings, content safeguards, and administrator features operate as designed.

3. Testing Scope

The testing scope includes registration and login, role-based access, creator dashboard, new-comic form, story settings, Gemini comic generation, response formatting, character and scene editing, panel editing, targeted regeneration, draft versions, preview modes, exports, project sharing, collaboration permissions, content reporting, administrator moderation, database operations, error handling, API credential protection, and responsive behavior on supported devices.

4. Test Environment

Testing should be completed in a development or staging environment separate from real production data. The environment includes the ComicCraft frontend, protected backend service, test database, configured Gemini credentials, supported web browsers, and representative mobile devices or emulators. Test accounts should be created for Creator and Administrator roles, with additional collaborator accounts where collaboration features are enabled. Test content should avoid unnecessary personal information and should not use real secrets.

5. Functional Testing

Functional testing checks whether each feature performs its intended task. A creator should be able to register, log in, create a comic project, enter story settings, generate an outline, edit characters and scenes, edit panels, save drafts, reopen a project, preview the comic, export the script, and change project privacy. An administrator should be able to manage authorized platform settings and review reports. Tests should include valid input, invalid input, missing required fields, and boundary cases.

6. Integration Testing

Integration testing confirms that ComicCraft modules work together. Important flows include frontend submission to backend validation, backend prompt construction to Gemini request, Gemini response to structured comic display, user edits to draft storage, saved project to preview and export, sharing invitation to collaborator access, reporting to moderation queue, and administrator action to content visibility update. A Gemini error or database failure should not corrupt the user interface or an existing saved draft.

7. Sample Test Cases

1. Valid creator login: Expected result – creator dashboard opens.
2. Empty story idea: Expected result – system requests a valid idea without calling Gemini.
3. Valid comic generation: Expected result – title, characters, outline, and panel script are displayed in the correct sections.
4. Save project: Expected result – draft is stored and visible only to the owner by default.
5. Edit one panel: Expected result – the selected panel updates without altering other panels.
6. Regenerate dialogue: Expected result – alternate dialogue is returned while the prior version can be retained.
7. Export comic: Expected result – selected project content is exported in the chosen format.
8. Share with viewer: Expected result – viewer can read but cannot edit the project.
9. Unauthorized direct project request: Expected result – access is denied.
10. Prohibited content prompt: Expected result – safe refusal or redirection is shown.
11. Gemini service failure: Expected result – friendly retry message appears and saved work remains available.

8. Usability and Accessibility Testing

Usability testing checks whether a first-time creator can understand the workflow from idea to export. Test users should locate Create New Comic, complete settings, edit a panel, save a draft, use preview, and export without unnecessary assistance. Accessibility checks include readable font size, sufficient contrast, labelled inputs, keyboard navigation where practical, clear form errors, usable buttons on touch screens, logical focus order, and responsive layouts for mobile, tablet, and desktop devices.

9. Content-Safety Validation

Safety testing checks that the application handles prohibited or sensitive requests according to the content policy. Test scenarios can include requests for disallowed sexual content, hateful content, harmful instructions, deceptive impersonation, or direct copyright imitation. The expected behavior is a safe refusal, a neutral explanation, or a suggestion to create original and appropriate content. Safety behavior should be checked across initial generation, targeted regeneration, dialogue assistance, and shared/public preview actions.

10. Security Validation

Security validation confirms authentication, session handling, password protection, authorization, private-project isolation, role permissions, secure API-key storage, input validation, output encoding, HTTPS configuration, rate limits, and administrative-access restrictions. Tests should verify that users cannot access another creator’s projects by changing a URL or API identifier. The Gemini credential must not appear in client code, public repositories, exports, or browser network responses.

11. Performance and Reliability Checks

Performance checks observe page load time, project-saving time, preview rendering, export preparation, and Gemini request response under typical use. The system should show loading states for longer operations and remain responsive during generation. Reliability checks include repeated generations, slow network conditions, session expiry, temporary API unavailability, malformed AI output, interrupted saving, and recovery after service availability returns. The user should not lose a saved draft due to a failed external request.

12. Defect Management

Each issue found during testing should be recorded with an identifier, module, test condition, expected result, actual result, severity, status, assigned person, and resolution notes. Security, privacy, data-loss, and core-generation failures are high priority. After a fix, the relevant test is rerun and regression checks are performed to ensure that the change has not affected unrelated features. Unresolved low-priority visual improvements may be documented for later versions.

13. User Acceptance Testing

User acceptance testing involves a small set of target users such as students, beginner creators, or content creators. They complete realistic tasks: generate a short comic, edit a character, change a panel, save a draft, preview the story, export it, and optionally share it. Feedback is collected on ease of use, usefulness of AI suggestions, clarity of safety messages, quality of generated structure, and confidence in saving or privacy controls. Feedback is used to make final corrections before deployment.

14. Acceptance Criteria

The phase is successful when critical creation flows pass; users can safely create, edit, save, preview, and export comics; private projects remain protected; role and collaboration permissions work correctly; content-safety handling is consistent; Gemini credentials remain secure; high-severity defects are resolved; and user acceptance feedback confirms that the application is understandable for its intended creators.

15. Phase Deliverables

The Phase 7 deliverables are the test plan, test cases, test environment setup, functional and integration test results, usability observations, accessibility checklist, content-safety validation evidence, security-validation checklist, performance observations, defect log, retest records, user acceptance feedback, and release-readiness summary.

16. Planned Next Phase

The next phase will be **Phase 8: Deployment, Final Documentation and Project Evaluation**. It will prepare ComicCraft for production or demonstration deployment, finalize user and technical documentation, create the project demonstration plan, evaluate results against objectives, and record future enhancements.

17. Conclusion

Phase 7 confirms that ComicCraft is tested as a complete AI-assisted comic-creation system rather than as isolated modules. Functional, usability, safety, privacy, performance, and security validation help ensure that creators can use the platform reliably and responsibly. After critical findings are addressed, ComicCraft is ready for final deployment and evaluation.

Phase Approval

Prepared by	_____

Guide / Faculty	_____
Approved by	_____